

# **Internationalize** your Symfony application, the right way





# Hello!

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SymfonyWorld  
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**DO YOU KNOW WHAT THEY CALL A  
QUARTER POUNDER WITH CHEESE IN PARIS?**

**A ROYALE CHEESE**

imgflip.com

Sorry for my french accent 🇫🇷



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## Why translation?

In 2016, a client in passenger transport domain needed a new European website

I have worked on this translation process



# Promise of this talk

Define a **process** and a **workflow** to ease the translation of your application, with **dedicated tools** for each person involved in the project.





## Some vocabulary

### i18n

#### Internationalization:

Designing a software that can be adapted to various languages without engineering changes.

### l10n

#### Localization:

Process of adapting an internationalized software for a specific region or language by translating texts, adapting date, currency formats.

### g11n

#### Globalization:

Combination of i18n and l10n process.



# Internationalize a Symfony Application

Why? When? How?



## Why?

- All your visitors don't speak your language
- Even if you target one country, all people in a country don't speak the same language (Switzerland, Belgium, ...)
- Centralize UI texts in files instead of spreading them in many templates.



## When?

- At the very beginning of the project
- Even for one language. Because centralizing UI texts in files will make your life easier later
- In an existing application is very expensive in time and money. And boring.

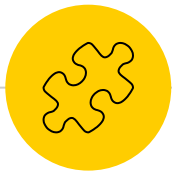


## How?

- ◉ Define a **clear process** with responsibilities for every involved workmate
- ◉ Use **dedicated tools** for developers, translators and QA
- ◉ **Document** the workflow and **train** your team to the tools.



# How does software i18n work?



HOW STANDARDS PROLIFERATE:  
(SEE: A/C CHARGERS, CHARACTER ENCODINGS, INSTANT MESSAGING, ETC.)



# Translation file format standards

.po, .mo, .ini, .yaml, .lang, .json, ...



**.po, .mo, .ini, .yaml, .lang, .json, ...**

- There are many language file formats
- But, there is **one standard**
- Defined by the OASIS\* consortium
- It is **XLIFF**

\*OASIS: Organization for the Advancement of Structured Information Standards



## XLIFF

- XML Localisation Interchange File Format
- 1.2 last revised 1<sup>st</sup> February 2008
- 2.0 last revised 5<sup>th</sup> August 2014



## XLIFF

- It is a **standard**, Symfony supports 1.2 and 2.0
- Has the best support in professional Translation Management Systems
- Symfony supports <notes> tag, useful to give more context to translation keys

# Use the XLIFF Format for **Your** Translation Files

“



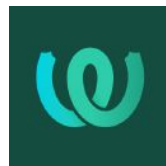
## Pick a tool

### Desktop



POEdit

### SaaS



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# Distribution of roles

Who's doing what?



## Assigning tasks

### Developers

- Define nomenclature for translation keys
- Create translation keys
- Use XLIFF notes tags to add context

### Translators

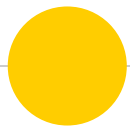
- Translate  
obviously
- Read notes tags provided by developers to understand context of keys

### Product Owners / QA

- Ensure that translations are ok in a staging environment
- Could mark translation as incorrect

```
<?xml version="1.0" encoding="utf-8"?>
<xliff xmlns="urn:oasis:names:tc:xliff:document:2.0" version="2.0"
srcLang="fr-FR" trgLang="fr-FR">
  <file id="messages.fr_FR">
    <unit id="IrB7E1R" name="user_account.last_login">
      <notes>
        <note>parameters:</note>
        <note>'formatted_datetime': '12 novembre 2020 à 19h22'</note>
      </notes>
      <segment>
        <source>user_account.last_login</source>
        <target>Dernière connexion le {formatted_datetime}</target>
      </segment>
    </unit>
  </file>
</xliff>
```





# Common mistakes

What should we avoid?



## A cumbersome process

- Translators don't have to edit XLIFF files.
- Code and text must not be mixed up
- Developers don't have to export new translations files manually



## Translation keys management

`user_profile.edit_form_part_1.fields.name.label` ❌

`form.label.name` ✅

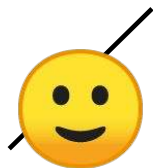
For ambiguous keys, use the <notes> tag of XLIFF.



## Store translations

### With code

In the translations directory of your application.  
Versioned with Git.



### SaaS

In a Translation Management System SaaS, like Loco or Crowdin.



**my\_app\_en\_GB\_v14\_2020\_03\_11\_0318pm.xlsx**

XLSX file, attached to an email conversation between developers and translators.



~~Hi developer, there's a typo on the  
homepage for title, could you  
please fix it and deploy it ASAP?~~

“

A product owner



## What do we have now?

### File format

XLIFF, full featured standard, supported by Symfony

### SaaS tool with an API

It allows developers to send new keys, and translators to translate them

### Keys nomenclature

Developers have guidelines to create translations keys

### Roles

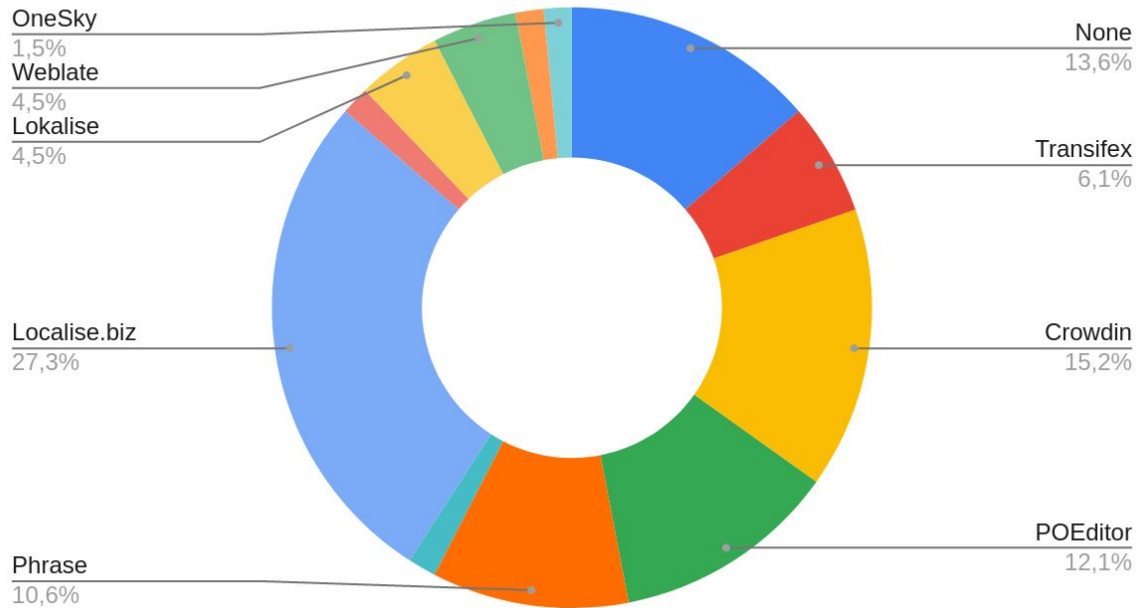
Everyone knows what they have to do



## I made a poll few weeks ago

Which Translation Management System do you use?

56 responses





## What do we miss?

Something to automate translations  
synchronization between SaaS API and all  
instances of your application.

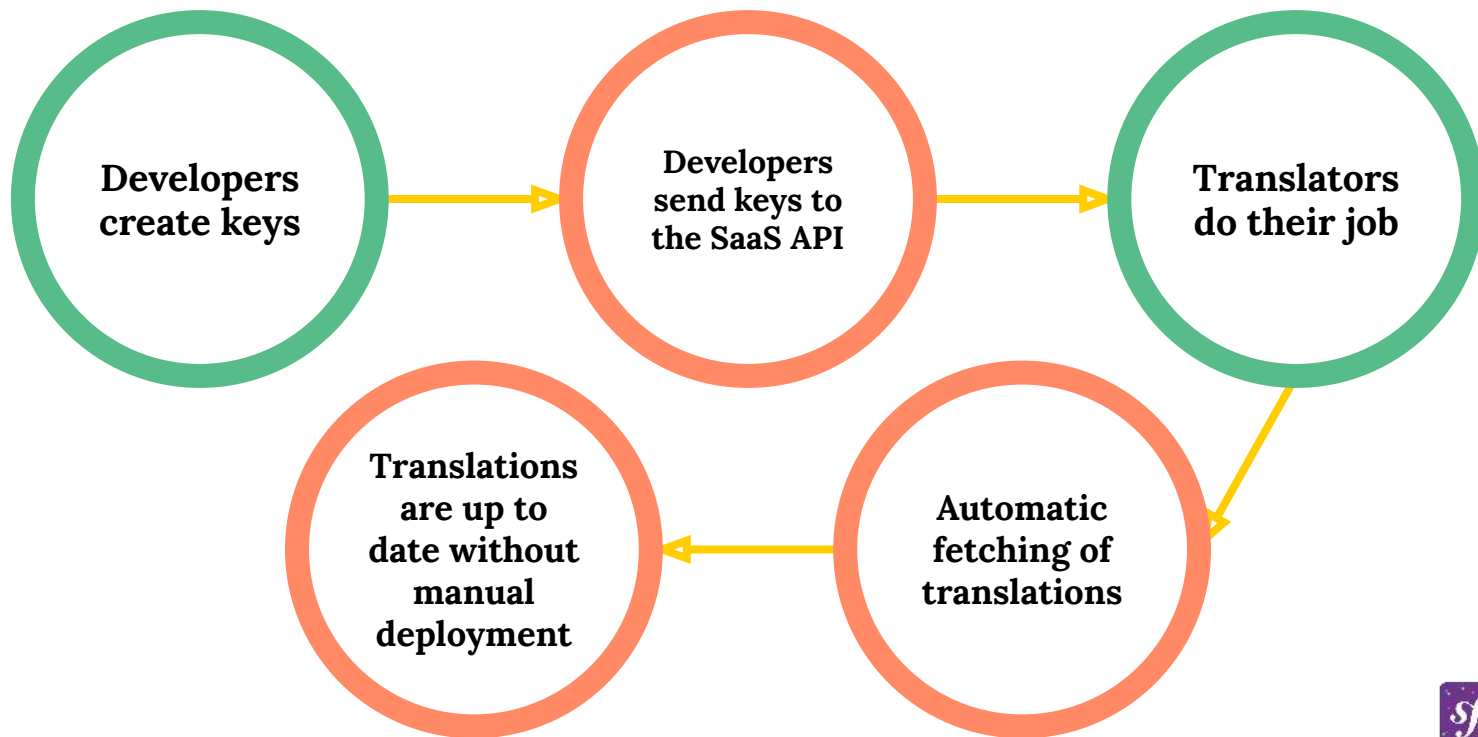


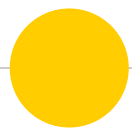
**Let's build the  
workflow!**





## Workflow





**<Code time!/>**



**Will we need a bundle?**

**YES**

But, which one?

# Compare bundles

|                                      | symfony/translation | JMSTranslationBundle | LexikTranslationBundle | php-translation/<br>symfony-bundle       |
|--------------------------------------|---------------------|----------------------|------------------------|------------------------------------------|
| Maintained                           | ✓                   | ✓                    | ✓                      | ✓                                        |
| Loaders<br>(read translation files)  | ✓                   | ✓ (homemade)         | ✗                      | ✓                                        |
| Dumpers<br>(write translation files) | ✓                   | ✓ (homemade)         | ✓ (only in database)   | ✓                                        |
| GUI                                  | ✗                   | ✓                    | ✓                      | ✓                                        |
| Domains management                   | ✓                   | ✓                    | ✓                      | ✓ (depends on SaaS)                      |
| SaaS API Clients                     | ✗                   | ✗                    | ✗                      | ✓                                        |
| Storages                             | Files               | Files                | Database               | Multiple (files,<br>SaaS, database, ...) |



*php-translation/symfony-bundle*  
WINS!

“

```
translation_adapter_loco:
  projects:
    my_application_messages:
      api_key: '%env(LOCO_MESSAGES_API_KEY)%'
      domains: ['messages']
    my_application_security:
      api_key: '%env(LOCO_SECURITY_API_KEY)%'
      domains: ['security']
```



```
translation:
  locales: ["en", "fr", "sv"]
  configs:
    app:
      dirs: ["%kernel.root_dir%/Resources/views", "%kernel.root_dir%/ ../src"]
      output_dir: "%kernel.root_dir%/Resources/translations"
      excluded_names: ["*TestCase.php", "*Test.php"]
      remote_storage: ["php_translation.adapter.loco"]
      output_format: "xlf"
```



```
*/15 * * * * bin/console translation:download --env=prod
```



## php-translation/symfony-bundle

This bundle allow you to:

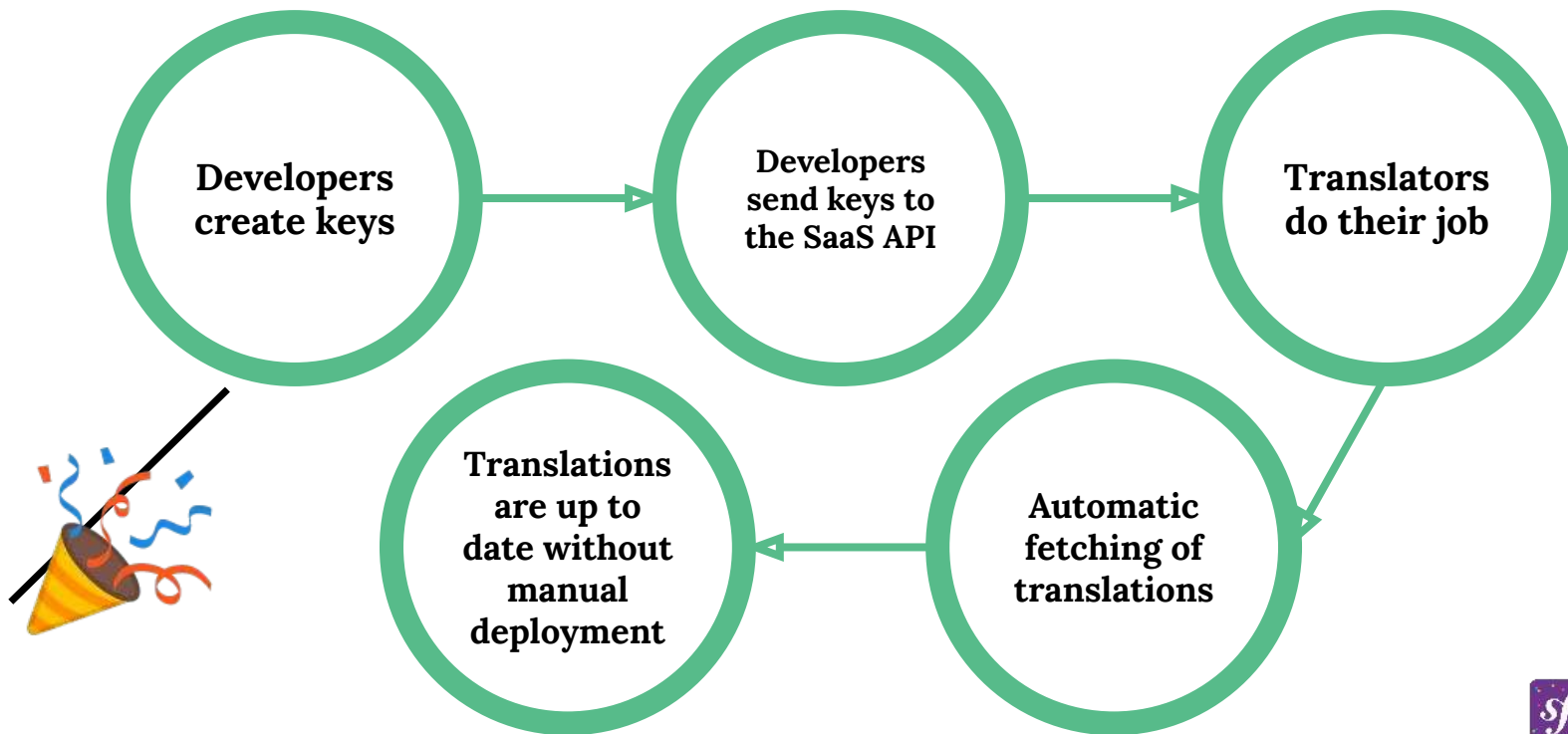
- Add new keys from the Symfony Profiler Translation panel
- Upload and download translations from SaaS to your application

| <input checked="" type="checkbox"/> | Locale | Domain   | Times used | Message ID                            |
|-------------------------------------|--------|----------|------------|---------------------------------------|
| <input checked="" type="checkbox"/> | fr     | messages | 1          | seo.default.keywords                  |
| <input checked="" type="checkbox"/> | fr     | messages | 1          | seo.default.robots                    |
| <input checked="" type="checkbox"/> | fr     | messages | 1          | seo.default.canonical                 |
| <input checked="" type="checkbox"/> | fr     | messages | 1          | search.form.passengers.baby_info.html |
| <input checked="" type="checkbox"/> | fr     | messages | 2          | booking.results.bundle.standard       |

Send selected translations to SaaS



## Workflow



*Oh, there's a typo on the  
homepage first title. Hurry up,  
let's fix it in {chosen SaaS}, and  
wait for the next translations  
fetch in X minutes*

“

*A product owner*



## Some ProTips<sup>©</sup>

### Write permissions

Ensure that your user has write permissions on translations directory.

### Be careful with crontab

Don't fetch your translation too often, it will clear translation cache each time.

### Phone number l10n

Use [odolbeau/phone-number-bundle](https://github.com/odolbeau/phone-number-bundle)

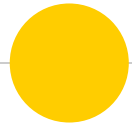
And format international phone numbers painlessly.

### Translations in test env?

Decorate the Symfony Translator, return the \$id except if \$domain == 'route'.

In your tests files, check if the keys are in your page.

One more thing...



**What if this workflow didn't  
require third party bundle?**

# [Translation][FrameworkBundle] Adding Translation Providers #38475

Edit Open with ▾

 Open welcoMatic wants to merge 14 commits into [symfony:5.x](#) from [php-translation:feature/remote-storages](#) 

 Conversation 0  Commits 14  Checks 4  Files changed 40

+3,051 -49 



welcoMatic commented on 7 Oct

Contributor 

| Q             | A                          |
|---------------|----------------------------|
| Branch?       | 5.x                        |
| Bug fix?      | no                         |
| New feature?  | yes                        |
| Deprecations? | no                         |
| Tickets       | <a href="#">Fix #36543</a> |
| License       | MIT                        |
| Doc PR        | todo                       |

Follow up of [#37462](#)

This PR refers to the early opened RFC [#36543](#) about adding third party translation SaaS into Symfony.  
I worked in collaboration with [@odolbeau](#) on this first draft.

We have implemented only Loco Provider for now, to validate the main workflow of the feature.  
We are not very sure about some naming convention, such as ~~RemoteStorage~~Provider, it might be renamed in Transport, to correspond to the naming of third party services in Mailer and Notifier Components.

This PR brings 2 new commands in FrameworkBundle: `translation:push` and `translation:pull`.  
It adds also new configuration entry:

```
framework:
    default_locale: fr
    translator:
```

## Reviewers

No reviews  
Still in progress? Convert to draft

## Assignees

No one assigned

## Labels

**Feature** Status: Needs Review Translator

## Projects

None yet

## Milestone

5.x

## Linked issues

Successfully merging this pull request may close these issues.

 [\[RFC\] \[Translation\] Remote Storages](#)

## Notifications

Customize

 Unsubscribe

[PR #38475](#) is open! 



**PR #38475 is open!**



It brings Translation Providers to Symfony.

It lets you use the Translation SaaS you want, use or write the Provider, and automate your translation workflow!



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**PR #38475 is open!**



You will be able to:

- Push new translations keys, remove obsolete ones to your chosen translation SaaS
- Pull new or updated translations
- Do both for each locale and/or domain





## Thanks a lot to

- Olivier Dolbeau to initiate this PR with me
- Tobias Nyholm and all contributors of php-translation
- JoliCode for sponsoring time to let me work on this PR
- Symfony team for selecting this talk



# Thanks!

*Any* **questions** ?

You can find me at

**@welcomattic** on Twitter and GitHub